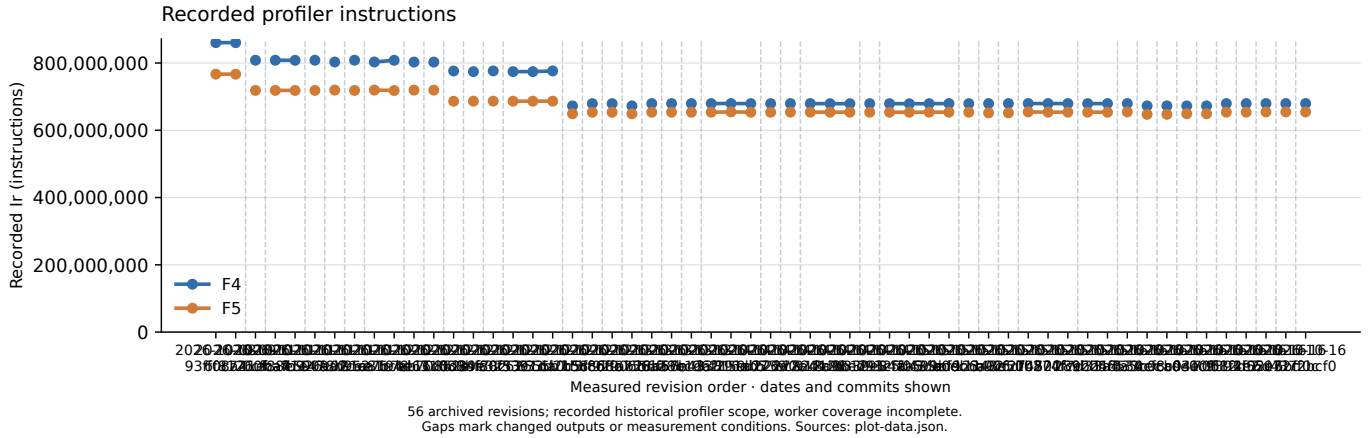
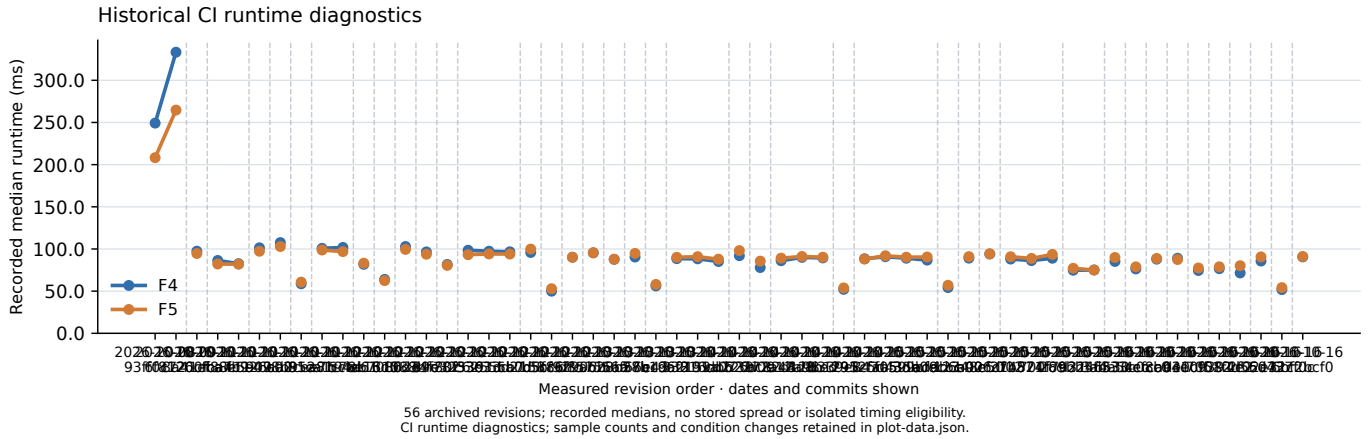
# Boolean matrix step: F4 and F5

One shared K_1 / 2^23 m=2 system, 22 Boolean variables, maximum degree 4. Matrix construction, reduction and readback; F5 includes its criterion.




| UTC date / revision | F4 Ir | F5 Ir | F4 ms | F5 ms |
|---|---|---|---|---|
| 2026-10-08 / 93ff0c72 | 860,633,889 | 766,940,469 | 249.241 | 208.250 |
| 2026-10-09 / 6f81260f | 860,555,567 | 766,776,087 | 333.279 | 264.687 |
| 2026-10-11 / 8a41eca7 | 808,263,705 | 718,576,235 | 97.380 | 94.658 |
| 2026-10-11 / 0cfba411 | 808,263,705 | 718,578,523 | 86.390 | 82.214 |
| 2026-10-11 / f88e5909 | 807,989,935 | 718,391,055 | 82.597 | 81.948 |
| 2026-10-11 / fb926c02 | 808,263,705 | 718,576,189 | 101.379 | 97.323 |
| 2026-10-11 / e478b952 | 803,017,657 | 719,516,063 | 107.585 | 102.927 |
| 2026-10-11 / aa0cee7b | 808,281,869 | 718,492,761 | 58.723 | 60.673 |
| 2026-10-12 / 81ca1c78 | 803,015,877 | 719,509,003 | 100.762 | 98.775 |
| 2026-10-12 / a8194d61 | 808,072,539 | 718,282,030 | 101.827 | 96.904 |
| 2026-10-12 / 5cbe16d5 | 802,824,065 | 719,316,911 | 81.784 | 83.144 |
| 2026-10-12 / eb70862e | 802,824,065 | 719,316,876 | 63.944 | 62.505 |
| 2026-10-12 / 31088bfc | 776,242,363 | 686,315,894 | 102.929 | 99.489 |
| 2026-10-12 / 0d34f782 | 774,379,185 | 686,521,131 | 96.521 | 93.790 |
| 2026-10-12 / 89e3233c | 776,490,367 | 686,543,371 | 81.628 | 80.581 |
| 2026-10-12 / 80f5391d | 774,379,185 | 686,519,896 | 98.501 | 93.239 |
| 2026-10-12 / c62555b7 | 774,379,185 | 686,519,896 | 97.360 | 94.117 |
| 2026-10-12 / 633d2158 | 776,467,358 | 686,512,000 | 96.637 | 94.008 |
| 2026-10-12 / ca0d5f86 | 672,500,776 | 649,174,334 | 95.848 | 99.895 |
| 2026-10-12 / b5cc6ffc | 679,242,389 | 653,622,849 | 49.965 | 52.882 |
| 2026-10-13 / bf92906b | 679,242,389 | 653,622,849 | 90.163 | 90.201 |
| 2026-10-13 / d85756aa | 672,486,486 | 649,160,075 | 95.497 | 95.480 |
| 2026-10-13 / ab28b57b | 679,449,939 | 653,823,121 | 87.697 | 87.878 |
| 2026-10-13 / 56f66e43 | 679,449,939 | 653,828,331 | 90.448 | 94.715 |
| 2026-10-13 / 28b1daf2 | 679,449,939 | 653,828,331 | 56.215 | 57.961 |
| 2026-10-14 / cc962159 | 679,449,939 | 653,831,656 | 88.546 | 90.149 |
| 2026-10-14 / 33816db5 | 679,549,997 | 654,627,509 | 88.344 | 90.874 |
| 2026-10-14 / 191a0739 | 679,240,609 | 653,664,759 | 85.135 | 87.917 |
| 2026-10-14 / cd226d0c | 679,240,609 | 653,664,759 | 92.059 | 98.073 |
| 2026-10-14 / b2272a44 | 679,449,939 | 653,873,401 | 78.007 | 85.621 |

| UTC date / revision | F4 Ir | F5 Ir | F4 ms | F5 ms |
|---|---|---|---|---|
| 2026-10-14 / bc82d128 | 679,449,939 | 653,873,401 | 86.126 | 89.102 |
| 2026-10-14 / 912a98ed | 679,098,623 | 653,830,968 | 89.995 | 91.125 |
| 2026-10-14 / 8d8b1295 | 679,098,623 | 653,830,968 | 89.395 | 90.230 |
| 2026-10-14 / b339e847 | 679,240,608 | 653,621,583 | 52.150 | 53.839 |
| 2026-10-15 / 7912faa5 | 679,226,318 | 653,606,027 | 88.406 | 88.091 |
| 2026-10-15 / 3c5d435e | 678,905,208 | 653,631,022 | 90.855 | 91.956 |
| 2026-10-15 / f0599bf9 | 679,098,622 | 653,825,706 | 89.060 | 90.253 |
| 2026-10-15 / 3db0ecbc | 679,449,938 | 653,830,913 | 86.813 | 90.492 |
| 2026-10-15 / addb6a02 | 679,256,525 | 653,632,330 | 54.248 | 56.951 |
| 2026-10-15 / 12348e50 | 679,449,939 | 651,714,930 | 89.150 | 90.941 |
| 2026-10-15 / b906df02 | 679,449,939 | 651,714,179 | 94.066 | 94.227 |
| 2026-10-15 / cf274521 | 679,549,984 | 654,627,503 | 88.061 | 90.795 |
| 2026-10-15 / 1b874fcc | 679,449,939 | 653,830,975 | 86.246 | 88.831 |
| 2026-10-15 / 700f7d31 | 679,444,579 | 653,824,679 | 89.023 | 93.545 |
| 2026-10-15 / 289bb5fb | 679,264,149 | 653,687,604 | 74.802 | 77.102 |
| 2026-10-16 / 92246530 | 679,449,938 | 653,829,966 | 75.128 | 75.009 |
| 2026-10-16 / 3a4a5cec | 679,549,995 | 654,621,347 | 85.229 | 89.944 |
| 2026-10-16 / 333c08c0 | 672,605,656 | 647,427,799 | 76.341 | 78.831 |
| 2026-10-16 / 4cfcb94e | 672,605,656 | 647,427,799 | 87.944 | 88.781 |
| 2026-10-16 / ca0310ff | 672,486,486 | 649,202,105 | 89.012 | 87.501 |
| 2026-10-16 / de0c9391 | 672,500,776 | 649,216,408 | 74.498 | 77.487 |
| 2026-10-16 / 7051f4e2 | 679,449,938 | 653,873,407 | 76.675 | 78.834 |
| 2026-10-16 / 082d5507 | 679,445,034 | 653,868,507 | 71.580 | 80.148 |
| 2026-10-16 / 2f02e32c | 679,549,984 | 654,627,503 | 85.711 | 90.622 |
| 2026-10-16 / 6d437f0c | 679,549,984 | 654,627,503 | 51.852 | 54.190 |
| 2026-10-16 / 6bf2bcf0 | 679,549,997 | 654,621,861 | 90.619 | 91.074 |

Historical Python-orchestrated perf.history/v1 snapshots; no new performance run or speedup claim.

Runtime: one recorded median per kernel and revision; sample counts retained per point, no stored spread, no isolated timing eligibility. Runtime is diagnostic.

Profiler Ir is the recorded historical scope. Worker-thread coverage is incomplete for these kernels; it is not a complete solver operation count.

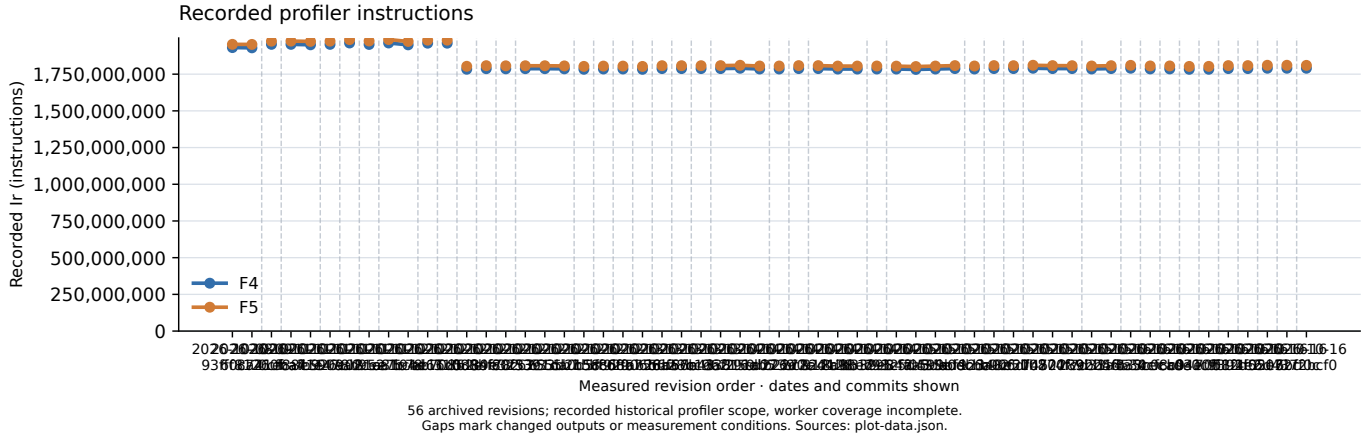
Lines stop at missing values, changed output fingerprints, or changed host/compiler/harness/profiler. Different algorithm fingerprints are expected and are not equated.

X positions are measured revision order, labelled with UTC commit date and revision; spacing is not elapsed time.

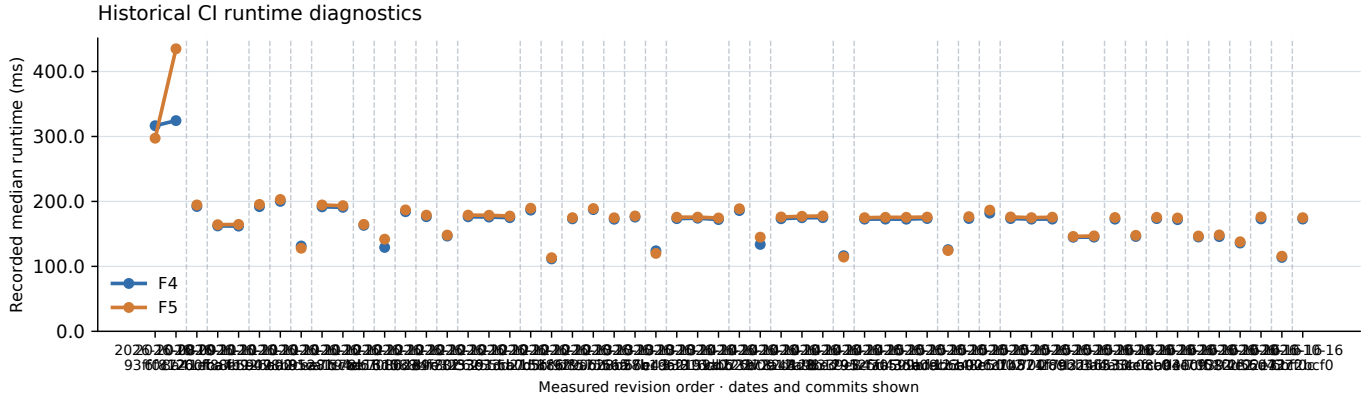
Evidence: 56 imported perf.history/v1 snapshots; historical_sources in plot-data.json retains raw filenames, SHA-256 hashes, hosts, harnesses and sample counts. Source data and provenance: plot-data.json and REPORT.md. These plots do not establish an end-to-end ECDLP gain.

# Boolean solver calls: F4 and F5

Shared K_1 / 2^17 m=2 workload, 16 Boolean variables, four planted and four random targets, maximum degree 3. The solve_boolean_system entry point, including root readback.




56 archived revisions; recorded historical profiler scope, worker coverage incomplete.
Gaps mark changed outputs or measurement conditions. Sources: plot-data.json.




56 archived revisions; recorded medians, no stored spread or isolated timing eligibility.
CI runtime diagnostics; sample counts and condition changes retained in plot-data.json.

| UTC date / revision | F4 Ir | F5 Ir | F4 ms | F5 ms |
|---|---|---|---|---|
| 2026-10-08 / 93ff0c72 | 1,930,989,200 | 1,953,108,992 | 316.570 | 297.297 |
| 2026-10-09 / 6f81260f | 1,929,374,570 | 1,952,598,745 | 324.403 | 435.004 |
| 2026-10-11 / 8a41eca7 | 1,953,021,023 | 1,974,789,730 | 192.203 | 194.561 |
| 2026-10-11 / 0cfba411 | 1,953,020,766 | 1,975,267,516 | 162.266 | 164.161 |
| 2026-10-11 / f88e5909 | 1,949,981,977 | 1,973,219,339 | 161.982 | 164.584 |
| 2026-10-11 / fb926c02 | 1,953,020,766 | 1,974,789,781 | 191.988 | 195.374 |
| 2026-10-11 / e478b952 | 1,962,376,386 | 1,985,422,672 | 200.058 | 202.996 |
| 2026-10-11 / aa0cee7b | 1,952,402,641 | 1,975,976,609 | 131.376 | 127.933 |
| 2026-10-12 / 81ca1c78 | 1,962,376,386 | 1,985,933,163 | 191.630 | 194.567 |
| 2026-10-12 / a8194d61 | 1,949,694,281 | 1,972,772,104 | 190.899 | 193.455 |
| 2026-10-12 / 5cbe16d5 | 1,961,124,313 | 1,981,687,200 | 163.135 | 164.619 |
| 2026-10-12 / eb70862e | 1,961,124,313 | 1,981,686,883 | 129.169 | 141.847 |
| 2026-10-12 / 31088bfc | 1,783,643,232 | 1,803,568,510 | 184.204 | 186.912 |
| 2026-10-12 / 0d34f782 | 1,787,314,371 | 1,807,797,314 | 176.437 | 178.824 |
| 2026-10-12 / 89e3233c | 1,786,351,448 | 1,806,842,750 | 146.640 | 148.159 |
| 2026-10-12 / 80f5391d | 1,787,314,371 | 1,807,244,283 | 176.353 | 178.878 |
| 2026-10-12 / c62555b7 | 1,787,314,371 | 1,807,244,283 | 175.899 | 178.689 |
| 2026-10-12 / 633d2158 | 1,786,234,678 | 1,806,168,901 | 174.990 | 177.494 |
| 2026-10-12 / ca0d5f86 | 1,783,289,117 | 1,802,994,551 | 186.580 | 189.492 |
| 2026-10-12 / b5cc6ffc | 1,785,257,434 | 1,804,956,249 | 111.497 | 113.333 |
| 2026-10-13 / bf92906b | 1,785,257,434 | 1,804,956,249 | 173.194 | 174.965 |
| 2026-10-13 / d85756aa | 1,783,003,746 | 1,802,715,389 | 187.409 | 189.039 |
| 2026-10-13 / ab28b57b | 1,787,965,641 | 1,807,670,514 | 172.579 | 174.785 |
| 2026-10-13 / 56f66e43 | 1,787,969,850 | 1,807,674,723 | 175.850 | 177.627 |
| 2026-10-13 / 28b1daf2 | 1,787,969,850 | 1,807,674,723 | 123.983 | 119.998 |
| 2026-10-14 / cc962159 | 1,787,965,641 | 1,808,168,023 | 173.510 | 175.690 |
| 2026-10-14 / 33816db5 | 1,790,194,799 | 1,810,456,555 | 174.166 | 175.821 |
| 2026-10-14 / 191a0739 | 1,785,256,826 | 1,805,459,208 | 172.076 | 174.475 |
| 2026-10-14 / cd226d0c | 1,785,256,826 | 1,805,459,208 | 185.930 | 188.735 |
| 2026-10-14 / b2272a44 | 1,787,965,186 | 1,808,230,025 | 133.809 | 144.964 |

| UTC date / revision | F4 Ir | F5 Ir | F4 ms | F5 ms |
|---|---|---|---|---|
| 2026-10-14 / bc82d128 | 1,787,965,186 | 1,808,230,025 | 173.338 | 176.053 |
| 2026-10-14 / 912a98ed | 1,784,912,427 | 1,805,177,722 | 174.838 | 177.121 |
| 2026-10-14 / 8d8b1295 | 1,784,912,427 | 1,805,177,722 | 175.021 | 177.613 |
| 2026-10-14 / b339e847 | 1,785,255,750 | 1,805,524,217 | 116.412 | 114.358 |
| 2026-10-15 / 7912faa5 | 1,784,970,955 | 1,804,687,873 | 172.752 | 174.937 |
| 2026-10-15 / 3c5d435e | 1,782,202,991 | 1,801,914,239 | 172.763 | 175.551 |
| 2026-10-15 / f0599bf9 | 1,784,990,781 | 1,805,255,060 | 172.667 | 175.531 |
| 2026-10-15 / 3db0ecbc | 1,787,964,110 | 1,808,232,577 | 173.668 | 175.897 |
| 2026-10-15 / addb6a02 | 1,785,268,499 | 1,805,533,338 | 125.807 | 124.378 |
| 2026-10-15 / 12348e50 | 1,788,056,225 | 1,808,318,933 | 173.678 | 176.547 |
| 2026-10-15 / b906df02 | 1,788,056,225 | 1,807,750,349 | 181.927 | 186.574 |
| 2026-10-15 / cf274521 | 1,790,285,878 | 1,810,547,645 | 173.613 | 176.055 |
| 2026-10-15 / 1b874fcc | 1,787,976,859 | 1,808,241,709 | 172.732 | 175.034 |
| 2026-10-15 / 700f7d31 | 1,787,976,859 | 1,807,671,995 | 172.808 | 175.713 |
| 2026-10-15 / 289bb5fb | 1,785,371,563 | 1,805,638,717 | 144.794 | 146.086 |
| 2026-10-16 / 92246530 | 1,788,058,429 | 1,807,748,501 | 144.989 | 146.889 |
| 2026-10-16 / 3a4a5cec | 1,790,297,890 | 1,809,989,439 | 172.711 | 175.194 |
| 2026-10-16 / 333c08c0 | 1,785,727,004 | 1,805,891,517 | 146.016 | 147.750 |
| 2026-10-16 / 4cfcb94e | 1,785,727,004 | 1,805,891,517 | 173.541 | 175.495 |
| 2026-10-16 / ca0310ff | 1,783,097,959 | 1,802,805,840 | 171.952 | 174.458 |
| 2026-10-16 / de0c9391 | 1,783,382,754 | 1,803,084,509 | 145.240 | 146.698 |
| 2026-10-16 / 7051f4e2 | 1,788,058,429 | 1,808,325,583 | 145.816 | 148.406 |
| 2026-10-16 / 082d5507 | 1,788,058,429 | 1,808,325,583 | 135.905 | 137.873 |
| 2026-10-16 / 2f02e32c | 1,790,289,796 | 1,810,551,563 | 173.111 | 176.139 |
| 2026-10-16 / 6d437f0c | 1,790,289,796 | 1,810,551,563 | 113.741 | 115.682 |
| 2026-10-16 / 6bf2bcf0 | 1,790,220,548 | 1,809,884,826 | 172.852 | 174.962 |

Historical Python-orchestrated perf.history/v1 snapshots; no new performance run or speedup claim.

Runtime: one recorded median per kernel and revision; sample counts retained per point, no stored spread, no isolated timing eligibility. Runtime is diagnostic.

Profiler Ir is the recorded historical scope. Worker-thread coverage is incomplete for these kernels; it is not a complete solver operation count.

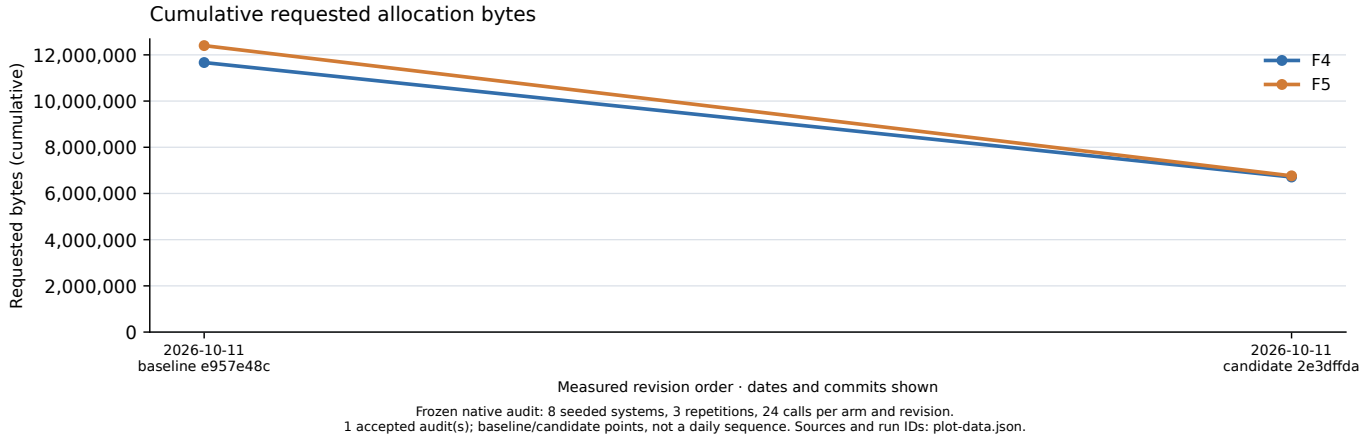
Lines stop at missing values, changed output fingerprints, or changed host/compiler/harness/profiler. Different algorithm fingerprints are expected and are not equated.

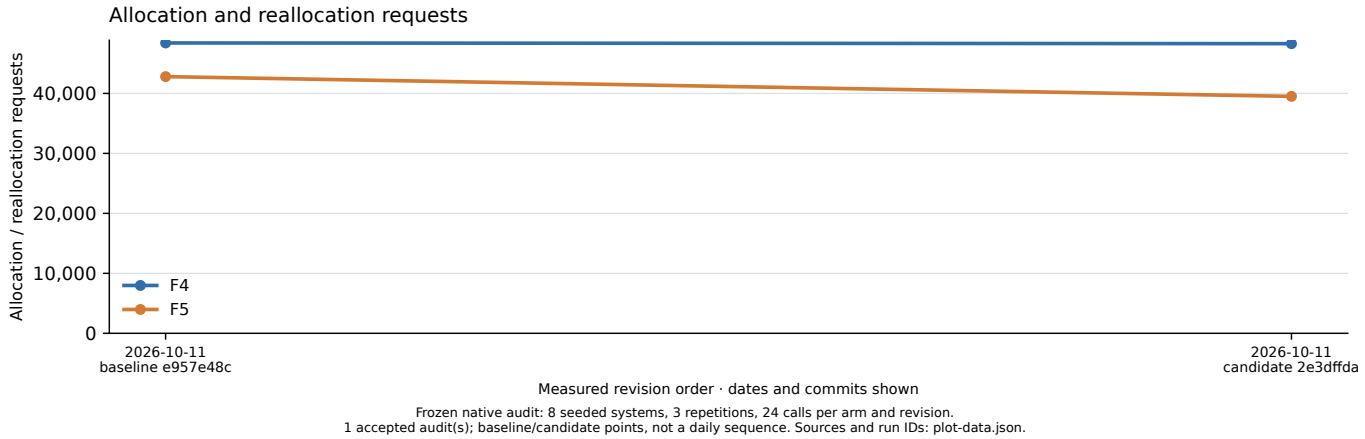
X positions are measured revision order, labelled with UTC commit date and revision; spacing is not elapsed time.

Evidence: 56 imported perf.history/v1 snapshots; historical_sources in plot-data.json retains raw filenames, SHA-256 hashes, hosts, harnesses and sample counts. Source data and provenance: plot-data.json and REPORT.md. These plots do not establish an end-to-end ECDLP gain.

## Complete Boolean calls: allocation history

Matching frozen Boolean allocation suite: 8 seeded systems, 3 repetitions, 24 calls per arm and revision. Polynomial dimensions, degrees and seeds are retained in the source audit protocol.




| UTC date / revision | F4 bytes | F5 bytes | F4 requests | F5 requests |
|---|---|---|---|---|
| 2026-10-11 / e957e48c | 11,667,696 | 12,400,920 | 48,414 | 42,801 |
| 2026-10-11 / 2e3dffda | 6,712,608 | 6,762,696 | 48,294 | 39,507 |

Native allocation audit. Original baseline/candidate rows, accepted comparison, exact outputs and accounting, shared inputs and all totals revalidated by the Rust plot-data tool.

One baseline and one candidate point per accepted audit, not a daily sequence. A/A and replication pairs are controls, not additional algorithm revisions.

Bytes are cumulative requested traffic, not peak memory. No timing is measured by this allocation series.

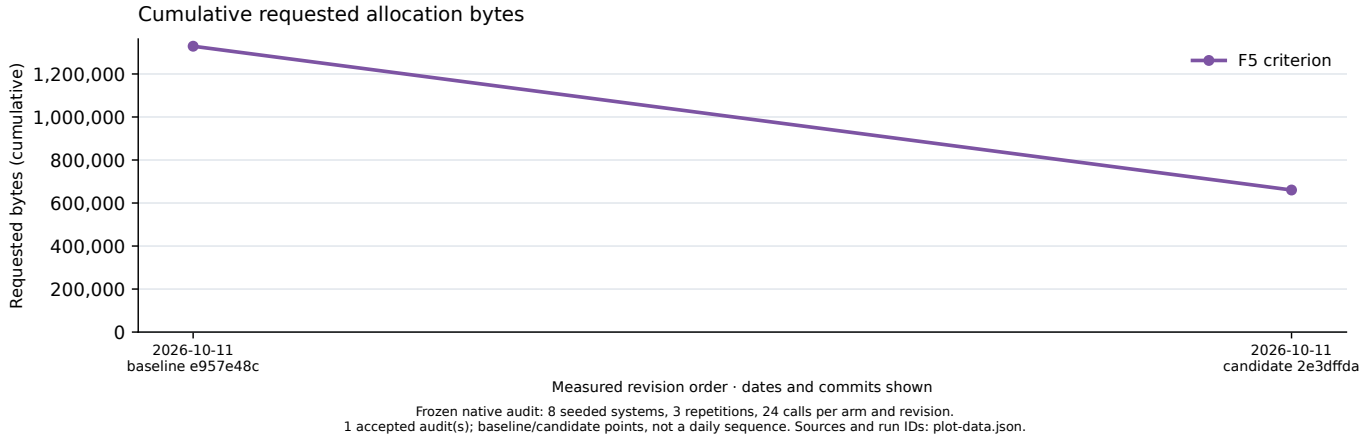
Criterion construction is a separate part of F5 and is never compared as a complete solver.

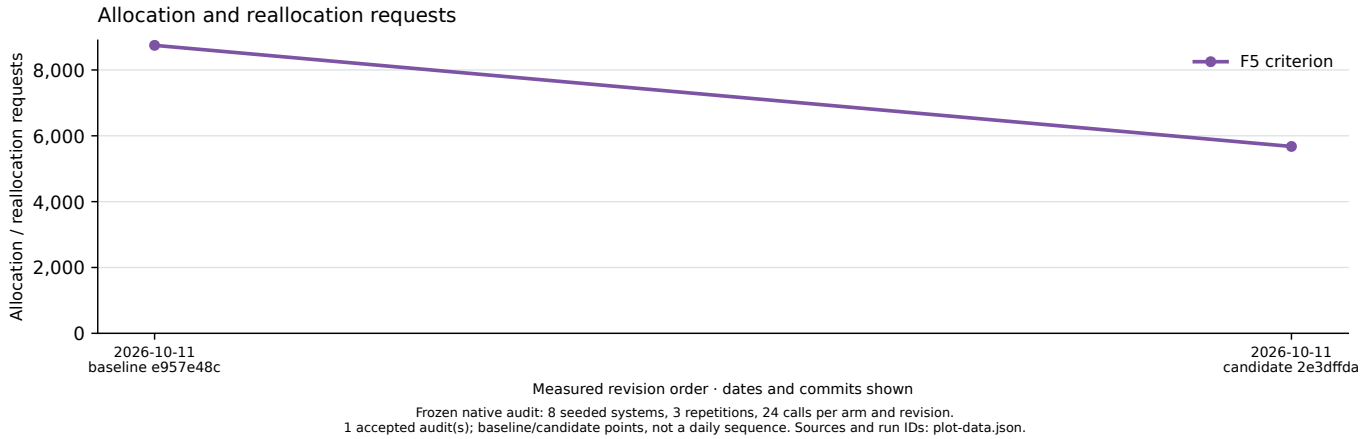
Audit revisions come from their frozen records. They are not inserted as measurements of historical CI workloads; the workloads and metrics remain separate.

Evidence: allocation_sources in plot-data.json; frozen native comparison and baseline/candidate JSONL records in research/f4_f5_row_build_20261011/records/. Source data and provenance: plot-data.json and REPORT.md. These plots do not establish an end-to-end ECDLP gain.

# F5 criterion: allocation history

Matching frozen Boolean allocation suite: 8 seeded systems, 3 repetitions, 24 calls per arm and revision. Polynomial dimensions, degrees and seeds are retained in the source audit protocol.




| UTC date / revision | F5 criterion bytes | F5 criterion requests |
|---|---|---|
| 2026-10-11 / e957e48c | 1,329,024 | 8,748 |
| 2026-10-11 / 2e3dffda | 660,480 | 5,676 |

Native allocation audit. Original baseline/candidate rows, accepted comparison, exact outputs and accounting, shared inputs and all totals revalidated by the Rust plot-data tool.

One baseline and one candidate point per accepted audit, not a daily sequence. A/A and replication pairs are controls, not additional algorithm revisions.

Bytes are cumulative requested traffic, not peak memory. No timing is measured by this allocation series.

Criterion construction is a separate part of F5 and is never compared as a complete solver.

Audit revisions come from their frozen records. They are not inserted as measurements of historical CI workloads; the workloads and metrics remain separate.

Evidence: allocation_sources in plot-data.json; frozen native comparison and baseline/candidate JSONL records in research/f4_f5_row_build_20261011/records/. Source data and provenance: plot-data.json and REPORT.md. These plots do not establish an end-to-end ECDLP gain.